Day 4 — SSL, High Availability, Self-Monitoring and Upgrades

Cisco AppDynamics Professional Implementer (500-430 CAPI) · 10-day course · Blueprint 2.2, 2.3, 2.4, 2.6 (Domain 2 = 25%)

Goal: secure the Controller with a CA certificate, design and run an HA pair, monitor the Controller itself, back it up, and upgrade the platform safely.
Verification: every fact checked against the current Splunk AppDynamics on-prem docs (October 2026).

Block 1 — Controller SSL (2.3)

Item	Default
HTTPS port	**8181**, self-signed certificate
Keystore	<controller_home>/appserver/jetty/etc/keystore.jks
Truststore	cacerts.jks (same folder) — CA root/intermediate certs, never private keys
Controller key alias	**s1as** — do not rename
Reporting Service alias	reporting-instance
Keystore password	**changeit** — well known, change it; keep keystore.jks and cacerts.jks on the same password

Replace the certificate (CSR flow)

```
cd <controller_home>/appserver/jetty/etc
cp keystore.jks keystore.jks.bak                        # 1. back up, stop the Controller
keytool -delete -alias s1as -keystore keystore.jks      # 2. remove self-signed key
keytool -genkeypair -alias s1as -keyalg RSA -keysize 2048 -validity 1825 \
        -keystore keystore.jks -ext SAN=dns:ctrl.bank.local   # 3. new key pair
keytool -certreq -alias s1as -keystore keystore.jks -file AppDynamics.csr   # 4. CSR
# 5. send CSR to the CA, receive signed certificate + chain
keytool -import -trustcacerts -alias s1as -file mycert.cer -keystore keystore.jks  # 6
# 7. restart the Controller
```

- **"Failed to establish chain from reply"** → import the root and intermediate CA certificates into **cacerts.jks**, then repeat step 6.
- **Existing key + certificate:** `openssl pkcs12 -export` → `keytool -importkeystore` → `-changealias` to **s1as**.
- **GUI:** Enterprise Console → Configurations → Controller Settings → Appserver Configurations → SSL Certificate Management → Edit Certificate.

HTTPS only

1. Stop the app server.
2. In `start.ini`, set `-Dappdynamics.controller.port` and `-Dappdynamics.controller.services.port` to the SSL port.
3. Add `-Dappdynamics.controller.ssl.enabled=true` so internal agents use SSL.
4. Change the deep-link URL to https.
5. Verify at `https://<host>:8181/controller`.

Agents over SSL (controller-info.xml)

```
<controller-port>8181</controller-port>   <!-- 443 via a proxy -->
<controller-ssl-enabled>true</controller-ssl-enabled>
<controller-keystore-password>...</controller-keystore-password>
<!-- only if the truststore is not the default cacerts.jks: -->
<controller-keystore-filename>../../conf/cacerts.jks</controller-keystore-filename>
```

Restart the JVM afterwards. A self-signed or private-CA certificate must be trusted by the agent's truststore.

Exam traps: s1as / changeit / 8181. If HTTPS is enabled, the Controller **hostname must not start with a digit** — upgrades fail (JDK limitation).

Block 2 — Reverse proxy and load balancer

- Agents and browsers use a **virtual IP** → move the Controller or switch the HA primary **without reconfiguring agents**. Adds a security layer (often in the DMZ).
- **SSL termination at the proxy** offloads SSL work from the Controller and keeps certificates in one place — recommended for large deployments.
- Set `-Dappdynamics.controller.ui.deeplink.url` to the external URL ending in /controller. If SSL ends at the proxy, also set `-Dappdynamics.controller.services.hostName` and `-Dappdynamics.controller.services.port` (usually 443). Edit via the EC Controller Settings page so changes persist.
- Proxy must send X-Forwarded-Proto: https and add Secure to cookies (X-CSRF-TOKEN isn't marked secure when SSL ends at the proxy).
- **.NET agents need a CA-signed certificate on the proxy** — self-signed won't work.
- Health check `/controller/rest/serverstatus`: **HTTP 200** on the active Controller, **HTTP 503 on the standby**.

Block 3 — High availability (2.4)

Primary + secondary Controller, **active-passive**, MySQL replicating primary → secondary. The load balancer / VIP sends traffic to **one** Controller only — never round-robin.

Prerequisite
Identical servers: OS, CPU, RAM, disk
Linux only for EC-managed HA
Passwordless SSH between hosts (RSA key pair + authorized_keys on each)
Both hostnames in each /etc/hosts; files before dns in /etc/nsswitch.conf
Two IPs per server (main + dedicated replication link) — recommended
Same database root password on both
C compiler (gcc) for the service installer

Setup

1. EC: add credentials and both hosts.
2. **Custom Install** with primary + **secondary** host (existing standalone Controller → **Add Secondary Controller** wizard).
3. `platform-admin.sh submit-job --service controller --job activate-ha-modules` — removes the EC from the failover path.

4. As root on both servers, from `<controller_home>/controller-ha`: `set_mysql_password_file.sh`, then `install-init.sh` (installs the appdcontroller and appdcontroller-db services).

Running HA

Task	How
Manual failover / failback	EC Controller page → HA Failover, or platform-admin.sh submit-job --service controller --job ha-failover --platform-name <name>
Automatic failover	The **Watchdog**. **Blocked** if the secondary database is down or replication is unhealthy
Restart primary without failover	Disable auto-failover first (EC Appserver Configurations → clear Enable Auto Failover, or --job stop-controller-watchdog); re-enable with start-controller-watchdog
Secondary > 3 days behind	--job incremental-replication (no downtime)
Secondary > 7 days behind	**Revive**: --job remove, then --job add-secondary — **restarts the primary (downtime)**

Block 4 — Self-monitoring (2.2)

- Bundled agent: `<controller_home>/appserver/jetty/appagent/javaagent.jar` — the **only** agent supported for self-monitoring.
- View: `/controller?enableAccounts=true` → account **system**, user **root**.
- Thread pools: Machine Agent reads JMX MBean domain `com.appdynamics.serverthreadpools`. JMX is on by default in Jetty; for a remote Machine Agent run `java -jar start.jar --add-to-start=jmx-remote` and restart.
- Disk latency: a SIM (Server Visibility) agent watches the data partition and alerts **above 3 ms**.
- HA best practice: agents on **both** Controllers report to a **separate monitoring Controller**.

Block 5 — Backup

Level	What	Suggested frequency
1 — Light	Controller install directory, excluding data (configuration)	Nightly
2 — Metadata	mysqldump-based script + keystores .appd.scskeystore and platform-admin/.appd.scs	Nightly
3 — Complete	Everything: metrics, snapshots, events, EC data	About weekly

- **Cold:** stop app server and database; copy install dir, MySQL datadir and JRE dir.
- **Hot:** Controller keeps running; Percona XtraBackup copies the datadir (can fail on very busy Controllers).
- **With HA:** stop the **secondary's** database, cold-backup there, restart it.
- **Exam fact:** the Enterprise Console **does not back up MySQL data** — back up yourself before upgrades.

Block 6 — Upgrades (2.6)

Order: Enterprise Console → Events Service → EUM Server → **Controller last**.

Before upgrading:

1. Read release notes for every version in between.
2. Check hardware and profile — profiles only grow.

3. Check db/logs/database.log for **InnoDB: Error** → if found, stop and contact support.
4. Record app server customizations (EC may not keep domain.xml, db.cnf changes).
5. New EC install? Create .passwordfile with AS_ADMIN_PASSWORD=<root password>.
6. HTTPS → hostname must not start with a digit.
7. **Back up** (Block 5).

- **Single Controller not yet managed by the EC:** Custom Install → **Discover & Upgrade** (discovery learns profile, tenancy, database settings), then upgrade.
- **HA pair (Upgrade Wizard):** upgrade EC → **fail over** to secondary → upgrade old primary (now standby) → **verify metrics in the last 5 minutes** (else "Failover & Rebuild Secondary" to roll back) → upgrade the other Controller. One Controller serves agents throughout.
- **Agents:** since 20.2, language agents are backward-compatible with any Controller.

Paper lab (~45 min)

1. **SSL:** write the keytool sequence to replace the self-signed certificate with a CA-signed one for ctrl.bank.local, and what to do on "Failed to establish chain from reply".
2. **HA design:** two Linux servers ctrl01 and ctrl02 behind VIP appd.bank.local — list prerequisites, setup commands and the load balancer health check.
3. **Maintenance:** OS patching needs the **primary** restarted without a failover — steps and commands.
4. **Upgrade plan:** EC, 3-node Events Service, EUM Server and an HA Controller pair — order, backups, Controller steps, verification, rollback.

Day 4 Quiz — with answer key

Cover the answer lines and test yourself first.

Q1. Which alias must the Controller's SSL key use in keystore.jks?

A. controller
B. s1as
C. reporting-instance
D. appd-ssl
Answer: B — s1as on port 8181 — don't rename it. reporting-instance is the Reporting Service.

Q2. Import gives 'Failed to establish chain from reply'. Fix?

A. Regenerate CSR with 4096-bit key
B. Import root + intermediate CA certs into cacerts.jks, then import again
C. Rename alias to controller
D. Change keystore password to changeit
Answer: B — keytool can't build the chain; add the CA certs to cacerts.jks and retry.

Q3. Behind a load balancer, what does /controller/rest/serverstatus return on the standby Controller?

A. HTTP 200 with available=true
B. HTTP 503
C. HTTP 401
D. A redirect to the primary
Answer: B — Active = 200, standby = 503, so the LB routes to the active node only.

Q4. Why run activate-ha-modules?

A. To start MySQL replication
B. To remove the Enterprise Console from the failover path
C. To license the secondary
D. To enable SSL between Controllers
Answer: B — Failover then works without depending on the EC.

Q5. Secondary database down for 10 days — what's required?

A. Nothing
B. Incremental replication only
C. Revive: remove, then add-secondary (restarts primary)
D. Reinstall the EC
Answer: C — More than 7 days behind → revive. Incremental replication is for more than 3 days.

Q6. When is automatic failover blocked?

A. Secondary DB down or replication unhealthy
B. More than 1,000 agents
C. Controller uses HTTPS
D. Events Service is a cluster
Answer: A — The Watchdog won't fail over to an unhealthy secondary.

Q7. Which Java Agent is supported for Controller self-monitoring?

A. Any version newer than the Controller
B. Only the bundled appserver/jetty/appagent/javaagent.jar
C. One in /opt/appdynamics/javaagent
D. The .NET Agent
Answer: B — Only the internal bundled agent is supported.

Q8. What does the EC back up automatically before a Controller upgrade?

A. Full MySQL datadir
B. Metadata tables only
C. Nothing — the EC does not back up MySQL data
D. Keystores only
Answer: C — You must back up yourself.

Q9. HA Upgrade Wizard — what happens after upgrading the EC and verifying credentials?

A. Both Controllers upgraded together
B. Fail over to secondary, then upgrade the old primary (now standby)
C. Uninstall the secondary
D. Upgrade the agents
Answer: B — Fail over, upgrade standby, verify metrics in last 5 minutes, then upgrade the other.

Q10. SSL terminated at a reverse proxy — which agents need a CA-signed certificate on the proxy?

A. Machine Agents
B. .NET agents
C. Database Agents
D. None
Answer: B — .NET agents need a CA-signed certificate; self-signed won't work.

Quick-review cheat sheet

Remember	Value
Controller SSL	8181 · keystore.jks · alias s1as · password changeit · truststore cacerts.jks
Chain error	Import root/intermediate into cacerts.jks, retry
HTTPS only	start.ini ports → SSL port · ssl.enabled=true · deeplink https
Hostname rule	Must not start with a digit when HTTPS is used
Proxy settings	deeplink.url · services.hostName/port (443) · .NET needs CA-signed cert
serverstatus	Active 200 · standby 503
HA basics	Active-passive · identical Linux servers · passwordless SSH · same DB root password
HA jobs	activate-ha-modules · ha-failover · stop/start-controller-watchdog
Replication lag	> 3 days incremental-replication · > 7 days revive (primary restarts)
Self-monitoring	Bundled appagent only · system account · JMX serverthreadpools
Backup	Levels 1/2/3 · cold vs hot (Percona) · HA → back up the secondary · EC doesn't back up MySQL
Upgrade order	EC → Events Service → EUM Server → Controller
HA upgrade	Fail over → upgrade standby → verify 5 min of metrics → upgrade other

Sources

- Controller SSL and Certificates
- Enable SSL between the Java Agent and the Controller
- Use a Reverse Proxy
- Set Up a High Availability Deployment
- Manage a High Availability Deployment
- Self-Monitoring the Controller
- Best Practices for Backups
- Back up the Existing Controller
- Before Upgrading
- Upgrade the HA Controller Pair Using the Upgrade Wizard
- Upgrade Platform Components
- Disk I/O Requirements

Paper lab — model answers

How an implementer would answer each Day 4 paper-lab task, with the rule behind each step.

1. Replace the Controller certificate for ctrl.bank.local

```
cd <controller_home>/appserver/jetty/etc
cp keystore.jks keystore.jks.bak
platform-admin.sh stop-controller-appserver
keytool -delete -alias s1as -keystore keystore.jks
keytool -genkeypair -alias s1as -keyalg RSA -keysize 2048 -validity 1825 \
        -keystore keystore.jks -ext SAN=dns:ctrl.bank.local
keytool -certreq -alias s1as -keystore keystore.jks -file AppDynamics.csr
#   → send AppDynamics.csr to the bank CA, receive mycert.cer + root/intermediate
keytool -import -trustcacerts -alias s1as -file mycert.cer -keystore keystore.jks
platform-admin.sh start-controller-appserver
```

If "Failed to establish chain from reply" appears:

```
keytool -import -trustcacerts -alias bankroot -file root.cer -keystore cacerts.jks
keytool -import -trustcacerts -alias bankinter -file intermediate.cer -keystore cacerts.jks
keytool -import -trustcacerts -alias s1as -file mycert.cer -keystore keystore.jks   # retry
```

- Keep the alias **s1as**; any alias works for CA certs in cacerts.jks.
- Replace the default password **changeit**, keeping keystore.jks and cacerts.jks on the same password.
- Verify at https://ctrl.bank.local:8181/controller. Because the bank CA is private, agents must trust it (import the root into the agent truststore).
- GUI alternative: EC → Controller Settings → Appserver Configurations → SSL Certificate Management.

2. HA design for ctrl01 / ctrl02 behind appd.bank.local

Prerequisites: identical Linux servers (OS, CPU, RAM, disk); passwordless SSH both ways for the Controller user; both hostnames in each /etc/hosts with files before dns in nsswitch.conf; same database root password; optional second NIC for replication; gcc installed.

Setup:

```
./platform-admin.sh add-credential --credential-name hakey --type ssh \
    --user-name appduser --ssh-key-file /home/appduser/.ssh/appd.pem
./platform-admin.sh add-hosts --hosts ctrl01 ctrl02 --credential hakey
# EC GUI → Custom Install: primary host ctrl01, secondary host ctrl02, profile, passwords
./platform-admin.sh submit-job --service controller --job activate-ha-modules
# as root on ctrl01 AND ctrl02, in <controller_home>/controller-ha:
./set_mysql_password_file.sh ...
./install-init.sh ...          # installs appdcontroller + appdcontroller-db services
```

- **Load balancer:** VIP appd.bank.local → pool ctrl01, ctrl02. Health check GET /controller/rest/serverstatus — healthy = HTTP 200 (active); the standby returns 503. Active-passive only, **no round-robin**.
- Agents use controller-host appd.bank.local (never ctrl01/ctrl02), so failover needs no agent change.
- If SSL ends on the LB: set deeplink.url to https://appd.bank.local/controller and services.hostName/port to appd.bank.local / 443.
- Add self-monitoring: agents on both Controllers report to a separate monitoring Controller.

3. Restart the primary for OS patching without failover

1. Disable auto-failover: EC → Appserver Configurations → clear **Enable Auto Failover** → Save, or `platform-admin.sh submit-job --job stop-controller-watchdog --service controller`.
2. On the primary: `bin/stopController.sh` (or `service appdcontroller stop`).
3. Patch and reboot the OS.
4. Start: `bin/startController.sh` (or `service appdcontroller start`); confirm serverstatus returns 200 through the VIP.
5. Re-enable auto-failover: `platform-admin.sh submit-job --job start-controller-watchdog --service controller` (or re-tick the box).

   Why: with auto-failover on, the Watchdog fails over if the app server is down more than 5 minutes. Alternative that avoids downtime: planned ha-failover to the secondary, patch the old primary, then fail back.

4. Full platform upgrade plan

Step	Action
0. Prepare	Read release notes for all versions in between; check OS/hardware support; check database.log for InnoDB: Error; record app server customizations; HTTPS hostname must not start with a digit
1. Back up	Controller: cold backup on the **secondary** (stop its DB, copy install dir, datadir, JRE, restart). Level 2 metadata (mysqldump + .appd.scskeystore, platform-admin/.appd.scs). EC data with mysqldump. The EC itself does not back up MySQL
2. Enterprise Console	Upgrade the EC first
3. Events Service	Upgrade the 3-node cluster from the EC Events Service page
4. EUM Server	Upgrade the EUM Server (back up its data first)
5. Controller HA pair	Upgrade Wizard: verify credentials → **fail over** to secondary → upgrade old primary (standby) → **verify**: log in, metrics in the last 5 minutes → upgrade the other Controller
Rollback	If verification fails: **Failover & Rebuild Secondary** — traffic returns to the not-yet-upgraded Controller
6. Post-checks	serverstatus 200 via VIP; agents reporting; EUM beacons arriving; Analytics searches return data; re-enable anything disabled
7. Agents	Upgrade later on their own schedule — agents are backward-compatible with any Controller (since 20.2)